

Introduction to Classification(ML)

Classification is a supervised learning technique in machine learning where the goal is to predict a category or label

Examples:

Spam/Not spam

Pass/Fail

Approved/Rejected loan

Logistic Regression:

supervised learning algorithm used for classification problems

used when the output variable is binary (two classes) such as:

Pass/Fail

0/1

Spam/Not spam

does not predict continuous values

instead, it predicts the probability of a class

example:

If we want to predict whether a student will pass:

input: study hours ,attendance ,marks

output: probability of passing (0 to 1)

Then we convert probability into class:

if probability >0.5 ---->Pass(1)

if probability <=0.5 ---->Fail(0)

Hypothesis Function:(Linear part)

first,logistics regression computes a linear combination of inputs:

$$z=wX+b$$

where:

* w=weight

* X=input feature

* b=bias

* z=linear output

Sigmoid function:

To convert the linear output into a probability,logistic Regression uses the sigmoid function.

Formula: $s(z)=1/1+e^{*-z}$

Properties:

* Output is always between 0 and 1

* Converts any real number into probability

* produces an S-Shaped (sigmoid) curve

Probability Interpretation:

After applying the sigmoid function,we get probability:

interpretation:

* if probability >=0.5 --->class=1

* if probability<0.5 ----> class=0

Decision boundary:

The decision boundary **is** the threshold that separates the two classes **in** logistic regression.

Condition: The decision boundary occurs when:

$$wX+b=0$$

Meaning:

* One side of the boundary ---->Class 0

* Other side of the boundary ---> Class 1

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
# Load dataset
df = pd.read_csv("student_data.csv")
# Features (X) and Target (y)
X = df[['Hours_Studied']]
y = df['Pass']
# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
# Create model
model = LogisticRegression()
# Train model
model.fit(X_train, y_train)
# Predictions
y_pred = model.predict(X_test)
# Predict new student
new_student = pd.DataFrame({
    'Hours_Studied': [6]
})
prediction = model.predict(new_student)
print(y_pred)
print("Prediction (0=Fail, 1=Pass):", prediction[0])

[1 0]
Prediction (0=Fail, 1=Pass): 1
```